Introduction

Matter is everywhere around us. The air we breath, water we drink and the material we use are made up of matter. Matter is defined as anything that occupies space and has mass. Matter is found in three major states: solid, liquid and gas. Do you know what is matter made of?

Matter is made of atoms. Atoms are the smallest particle of matter. They are so small that you cannot see them with your eyes or even with a standard microscope. A standard sheet of paper is about millions of atoms thick. Science has come up with a technology to identify the structure of atoms by using Atomic resolution Microscope (ARM) and Tunnelling Electron Microscope (TEM) which use electricity to map atoms. There is more about atoms in the later classes. But first let's learn about the three states of matter.

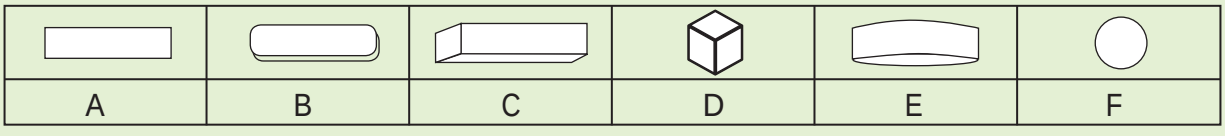
3.1 Physical Nature of Matter

Matter occupies space and has mass. What is its nature? Many philosophers pondered over this question and came out with ideas. It is known that Indian Philosopher **Kanada** and Greek philosopher **Democritus** had their ideas similar. The Indian philosopher Kanada called it as **paramanu** and Democritus called it as **atomos**.

Imagine that a piece of thread is cut endlessly using knife. At one point it would be like a small piece that it cannot be further cut by a knife. That small particle may contain millions of molecules and these molecules are made of atoms. Matter is made of such smallest particles 'atoms'. These atoms are extremely small even to see under a powerful microscope.

Activity - 1

Take a few crystals of sugar. Observe them carefully with the help of a magnifying lens.

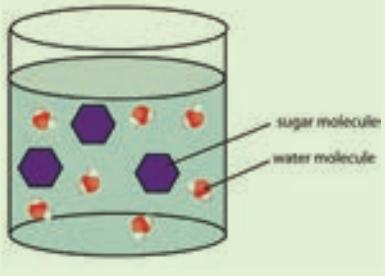


Which of the shapes given above resemble a sugar crystal?

A B C D E F

Now place a few sugar crystals into water.

What happens to the sugar crystals?



A sugar crystal is also made up of molecules. When sugar dissolves in water, the sugar crystals break down and the molecules of sugar get distributed in water. This makes water sweet in taste. The sugar molecules are extremely small; that is why we are not able to see them. Small amount of matter has many millions of molecules in it (1 million = 10 lakhs).